RES POLYTECHNIC, BANGALORE-91

CIE Practice Test IV

Program	Mechanical Engineering	**Semester - III**
Course Name	Programming Concepts in Mechanical Engineering	**Test IV**
Course Code	25ME34I	**Marks - 50**
Name of the Course Coordinator	Vinay Kumar	**Duration - 180 min**

Note: Answer any one full question from the following.

Q. No	Questions	CO	Marks
1	a. Write a Python program to calculate the mean, variance and standard deviation of the shaft diameters 24.9, 25.1, 25.0, 25.2, 24.8, 25.0, 25.1, 24.9, 25.0 and 25.0 mm.	CO2	5+5+10+ 15+15 = **50 Marks**
	b. The sample averages of 7 samples are 10.02, 10.05, 9.98, 10.10, 10.01, 9.90 and 10.03 mm, with UCL = 10.08 mm and LCL = 9.92 mm. Write a Python program to identify the in-control and out-of-control samples.	CO3	
	c. Develop a Python program to calculate the sample average (X-bar) and sample range (R) for the given data and plot the X-bar and R charts using Matplotlib.	CO3	
	d. Write a Python program to control a DC motor (forward, reverse and stop) using an L298N motor driver with Raspberry Pi.	CO4	
	e. Write a Python code to plot the stress distribution in the plate at various lengths along the bar.	CO3	
2	a. Write a Python program to calculate the process capability Cp and the process capability index Cpk for a process with mean = 50.02 mm, standard deviation = 0.15 mm, USL = 50.5 mm and LSL = 49.5 mm.	CO2	5+5+10+ 15+15 = **50 Marks**
	b. Write a Python program to calculate the sample average and the sample range of each of the five given samples (each of size four).	CO3	
	c. Generate Python code to plot the quality control tools (Pareto chart and histogram) for the given data.	CO3	
	d. Write a Python program to build an automated water pump controller using Raspberry Pi: the pump is switched ON when the tank level sensor reads LOW and OFF when it reads FULL.	CO4	
	e. Write a Python code to compare and plot the stress distribution in the plate with and without hole.	CO3	

Signature of the Course Coordinator

Signature of the HOD

Signature of the IQAC Chairman